

Vivek Patel

Sr. RF Design and Optimization Engineer - Mobility Inc

Honolulu, HI 96808

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Willing to relocate: Anywhere

Work Experience

Sr. RF Design and Optimization Engineer

Mobility Inc - Honolulu, HI

April 2019 to Present

- Drive data analysis in Actix and parameter optimization in Ericson OSS platform for LTE pre and post launch optimization.
- Diagnosed all RF related problems such as the drop in accessibility, retainability, mobility, throughput, drop call, block call and call setup failure, missing neighbor and Swap sector and SC consistency.
- Identified cell sites negatively impacted by uplink (UL RSSI) interference, implemented corrective measures such as checking and ensuring that tower mounted amplifiers(TMA) parameter are properly set.
- Physical optimization include change of power, azimuth, rad center, missing neighbors, mechanical tilt and electrical tilt changes.
- Parameter optimization using Moshell Commands from Ericsson OSS Command line interface.
- Troubleshooting RF issues (Pilot pollution-dominance, link imbalance, PCI Planning, Missing Neighbors), Idle mode Issues (access and paging, RRC Setup failures and RAB Set Up) using Truncal, Spectrum Analyzer and nGeniusONE.
- Ran the coverage prediction in Asset for various scenarios such as Antenna model, Tilts, RAD center, Azimuth, and coordinates.
- Adding/Deleting a neighbor Intra-frequency, Inter-frequency and IRAT Neighbors using Ericsson OSS.
- Draw the drive routes for cluster drive in MapInfo.

RF Design Engineer

AT&T - King of Prussia, PA

March 2014 to April 2019

- Worked in LTE cluster design using Atoll and ACP and got it approved from HQ.
- Designed TA packages for more than 100 Infill sites and got it approved from HQ.
- Greenfield, expansion and overlay LTE cluster design was completed.
- Traffic spreading was created for both data and voice and run using Monte Carlo simulations. Parameter settings were used according to the guidelines.
- Design optimization was carried out using electrical/mechanical down tilts and azimuth for KPI optimization such as SINR, RSRP, RSRQ, Number of server in 5 dB margin, Handover status, etc.
- Created search areas and reviewed search area candidates within geographic area by taking into consideration performance issues, capacity, topography, and population by using Atoll and MapInfo.
- Closeout packages include neighbor list creation. Scrambling codes were also created along with presentation for several clusters. PICD presentation was done after each Atoll project completion.

- Generated SARF, SCIP and CIQ forms as per RF market request.
- Reviewing and approving the Construction drawing (CD).

RF Optimization Engineer

MobileComm Professionals Inc - Atlanta, GA

December 2011 to March 2014

- Performed advanced monitoring and review of radio network Key Performance Indicators (KPI).
- Analyzed and optimized data using Actix, Ericsson OSS and made necessary changes to make all the KPIs green.
- Optimize the coverage, Ec/Io, drop call, block call and call setup failure, missing neighbor and swap sector and SC consistency.
- Drive data analyzing then optimizing activities include change of frequency, change of power, azimuth, rad center, missing neighbors, mechanical tilt and electrical tilt changes in Ericsson OSS.
- Identified pilot pollution and took corrective action such as reducing CPICH, performing foot print optimization to create dominant servers and requesting scrambling code retune.
- Candidate finalizing using coverage predication tool - ASSET.
- Updated the ASSET database from other RF database.
- Creating, updating, and deleting the RFDS for various project and carrier types.
- Responsible for redesign the RFDS if the structural analysis fails also responsible for ASSET updates.
- Created FCC and FAA report as per RF market request.
- Reviewing and approving the Construction drawing (CD).

RF Associate Engineer

MobileComm Professionals Inc - Atlanta, GA

June 2011 to December 2011

- Conducted drive testing and collected test data using Agilent and TEMS for LTE, UMTS and GSM technologies.
- Verify LTE/UMTS /UTRAN basic functionality.
- Collect UE physical measurement for the entire active cell.
- Verify handover functionality. Mobility test soft and softer handover between neighboring site and within the sector respectively.
- Drive test analysis of 4G/3G coverage and quality, call event analysis, handover analysis, swap feeder and missing neighbor analysis.
- Conducted IRAT testing (Functional and Boundary) on UMTS boundaries to check handoffs between UMTS and GSM.
- Perform single site verification (SSV) test for various cell sites on pre-launch of new 4G/3G cell sites.
- Identify major call related problems (dropped calls, hardware problems etc.)

Education

Bachelor of Science

Gujarat University - Gujarat, IN

Skills

ANALYZER, SPECTRUM ANALYZER, MAPINFO, VISIO, SQL